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Evaluating Alert Fatigue Codebook

Code

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Dataset Overview

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Table 1. Dataset Summary

Data Source	MIMIC-IV (PhysioNet / Beth Israel Deaconess Medical Center)
Unit of Analysis	ICU stay (stay_id), linked to hospital admission (hadm_id)
Total Observations	94,444
Total Variables	28
Date Constructed	March 2026
Software	R 4.x with data.table, dplyr, MASS, DuckDB, lubridate

Variable Codebook

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Table 2. Variable Codebook — All 27 Variables

Variable	Type	Source	Description

Variable	Type	Source	Description
subject_id	Integer	MIMIC-IV: icustays / patients / admissions	Unique patient identifier
hadm_id	Integer	MIMIC-IV: icustays / patients / admissions	Hospital admission ID; primary join key across tables
stay_id	Integer	MIMIC-IV: icustays / patients / admissions	ICU stay identifier
anchor_age	Integer	MIMIC-IV: patients	Patient age at MIMIC-IV anchor year
gender	Factor	MIMIC-IV: patients	Patient sex (M = Male, F = Female)
first_careunit	Factor	MIMIC-IV: icustays	ICU unit type at time of admission
admittime	POSIXct	MIMIC-IV: admissions	Hospital admission timestamp
dischtime	POSIXct	MIMIC-IV: admissions	Hospital discharge timestamp
intime	POSIXct	MIMIC-IV: icustays	ICU admission timestamp
outtime	POSIXct	MIMIC-IV: icustays	ICU discharge timestamp
hospital_los_days	Numeric	Derived	Hospital length of stay in days (dischtime - admittime)
icu_los_hours	Numeric	Derived	ICU length of stay in hours (outtime - intime)
icu_los_days	Numeric	Derived	ICU length of stay in days (icu_los_hours / 24)
n_prescriptions	Integer	MIMIC-IV: prescriptions (aggregated)	Total medication prescription orders per hospital admission
n_unique_drugs	Integer	MIMIC-IV: prescriptions (aggregated)	Count of distinct drug names ordered per hospital admission
rx_per_day	Numeric	Derived	Medication orders per ICU day (n_prescriptions / icu_los_days)
n_lab_events	Integer	MIMIC-IV: labevents (aggregated)	Total laboratory measurements per hospital admission
n_unique_labitems	Integer	MIMIC-IV: labevents (aggregated)	Count of distinct laboratory item types per admission
n_abnormal_labs	Integer	MIMIC-IV: labevents (aggregated)	Laboratory measurements flagged as abnormal
labs_per_day	Numeric	Derived	Lab events per ICU day (n_lab_events / icu_los_days)
total_events	Integer	Derived	Total CDS-triggering events (n_prescriptions + n_lab_events)
total_events_per_day	Numeric	Derived	Total events per ICU day (total_events / icu_los_days)
scenario1_events	Integer	Simulated (scenario analysis)	Simulated burden: Rx orders reduced 15% (deduplication scenario)
scenario2_events	Integer	Simulated (scenario analysis)	Simulated burden: Lab events reduced 20% (suppression scenario)
scenario3_events	Integer	Simulated (scenario analysis)	Simulated burden: Combined Rx -15% and Lab -20%
max_burst_60min	Integer	Derived: prescriptions timestamps	Maximum medication orders in any 60-minute window per admission

Variable	Type	Source	Description
burst_category	Factor	Derived: prescriptions timestamps	Burst intensity category: Low / Moderate / High / Very High

Descriptive Statistics

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Table 3. Descriptive Statistics: Burden and Temporal Variables

Variable	N	Mean	SD	Median	Q1	Q3	Min	Max
n_prescriptions	94444	113.8	116.2	83.0	48.0	136.0	0	2812.0
n_unique_drugs	94444	43.0	22.0	40.0	27.0	55.0	0	196.0
rx_per_day	94444	77.6	1179.6	35.8	20.9	64.7	0	297852.6
n_lab_events	94444	380.6	669.4	198.0	32.0	429.0	0	19824.0
n_unique_labitems	94444	53.8	40.9	54.0	23.0	81.0	0	267.0
n_abnormal_labs	94444	150.6	298.5	65.0	6.0	159.0	0	8421.0
total_events	94444	494.4	750.7	280.0	127.0	549.0	0	21980.0
total_events_per_day	94444	330.8	7632.1	126.9	64.9	244.5	0	1791094.7
max_burst_60min	94156	22.8	16.8	20.0	12.0	30.0	1	432.0
icu_los_days	94444	3.6	5.4	2.0	1.1	3.9	0	226.4
anchor_age	94444	63.0	16.7	65.0	53.0	76.0	18	91.0

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Table 3a. Gender Distribution

Gender	N	%
F	41577	44
M	52867	56

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Table 3b. ICU Length of Stay Summary (days)

Median LOS	Mean LOS	Q1	Q3	Max
2	3.6	1.1	3.9	226.4

Statistical Approach

Why Negative Binomial Regression?

The outcome variables in this analysis are counts — whole numbers that can never be negative (medication orders, lab tests, total events). Standard linear regression is not appropriate because it can predict impossible negative values and assumes a normal distribution that count data does not follow.

The simpler count model — Poisson regression — also does not fit here because it requires the variance to equal the mean. Our data is far more spread out than Poisson can handle. This is called **overdispersion**.

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Table 4. Overdispersion Check.

Variable	Mean	Variance	Variance / Mean
Medication orders	113.8	13491.3	**119**
Lab events	380.6	448148.2	**1177**
Total events	494.4	563618.3	**1140**

Negative binomial regression handles overdispersion by adding a dispersion parameter (theta) that absorbs the extra spread in the data. This produces honest standard errors and reliable confidence intervals.

Formal Confirmation: Poisson vs. Negative Binomial

Show

Table 5. Likelihood Ratio Test.

Comparison	Chi-Squared	p-value	NB Theta	Decision
Poisson vs. Negative Binomial	8,224,010	0e+00	1.063	**Negative Binomial**

What the Offset Term Does

All models include `offset(log(icu_los_days))`. Without this, a 30-day stay would appear to have higher burden than a 2-day stay simply because of duration, not intensity. The offset converts the model to predict **events per ICU day**, making every encounter comparable regardless of how long the stay was.

How to Read the Results (IRR)

Table 6. How to Read an Incidence Rate Ratio (IRR).

IRR	What It Means	From Our Study
1.0	No difference from the reference group	CVICU — our reference unit in Models 1 and 3
1.20	20% MORE events per day than the reference group	MICU/SICU vs. CVICU
0.81	19% FEWER events per day than the reference group	Example only
1.035	3.5% more events per day — small but real	Male vs. Female patients
0.189	81% FEWER events per day — very large difference	Neuro Intermediate vs. CVICU

Key Results: Forest Plot

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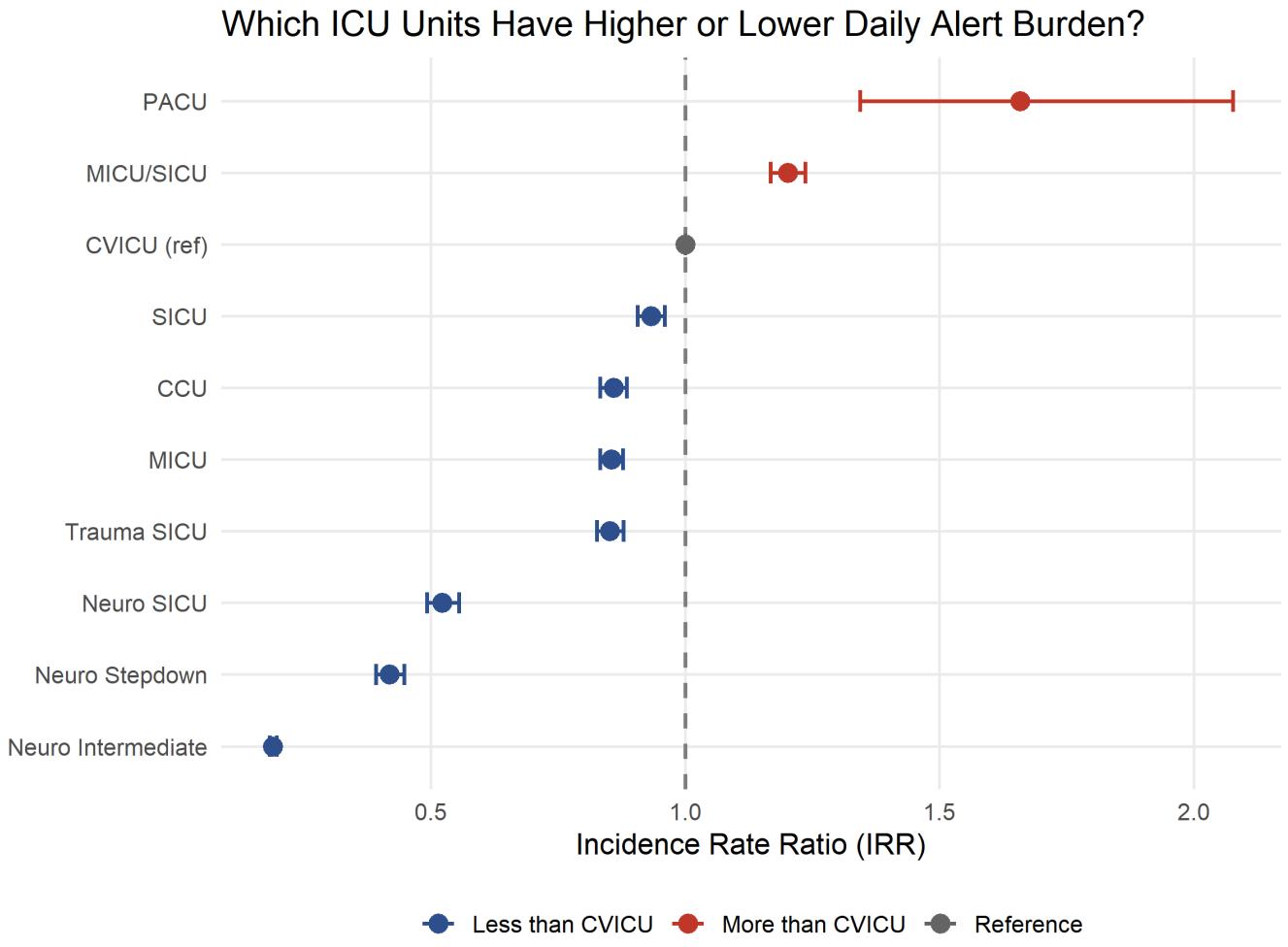

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Table 7. Model 1 — Medication Orders per Day

Predictor	IRR	Std. Error	Statistic	p-value	95% CI Low	95% CI High

Predictor	IRR	Std. Error	Statistic	p-value	95% CI Low	95% CI High
(Intercept)	89.856	0.016	283.156	0.0000	87.101	92.697
anchor_age	0.998	0.000	-12.818	0.0000	0.997	0.998
genderM	1.006	0.006	0.919	0.3582	0.993	1.019
first_careunitCoronary Care Unit (CCU)	0.885	0.012	-9.894	0.0000	0.864	0.906
first_careunitIntensive Care Unit (ICU)	0.640	0.170	-2.627	0.0086	0.459	0.893
first_careunitMed/Surg	0.584	0.978	-0.550	0.5825	0.086	3.974
first_careunitMedical Intensive Care Unit (MICU)	0.816	0.011	-19.195	0.0000	0.799	0.833
first_careunitMedical/Surgical Intensive Care Unit (MICU/SICU)	0.941	0.011	-5.388	0.0000	0.920	0.962
first_careunitMedicine	0.297	0.243	-4.996	0.0000	0.184	0.478
first_careunitMedicine/Cardiology Intermediate	3.601	0.971	1.320	0.1868	0.537	24.133
first_careunitNeuro Intermediate	0.310	0.015	-76.794	0.0000	0.300	0.319
first_careunitNeuro Stepdown	0.351	0.027	-38.339	0.0000	0.333	0.370
first_careunitNeuro Surgical Intensive Care Unit (Neuro SICU)	0.631	0.025	-18.623	0.0000	0.601	0.662
first_careunitNeurology	0.110	0.972	-2.270	0.0232	0.016	0.740
first_careunitPACU	7.205	0.088	22.348	0.0000	6.059	8.567
first_careunitSurgery/Trauma	0.426	0.308	-2.776	0.0055	0.233	0.778
first_careunitSurgery/Vascular/Intermediate	0.423	0.081	-10.596	0.0000	0.361	0.496
first_careunitSurgical Intensive Care Unit (SICU)	0.853	0.012	-13.466	0.0000	0.834	0.873
first_careunitTrauma SICU (TSICU)	1.008	0.013	0.610	0.5418	0.983	1.033

Show

Table 8. Model 2 — Laboratory Events per Day

Predictor	IRR	Std. Error	Statistic	p-value	95% CI Low	95% CI High
(Intercept)	256.469	0.032	173.658	0.0000	240.905	273.039
anchor_age	0.999	0.000	-3.561	0.0004	0.998	0.999
genderM	1.015	0.013	1.138	0.2550	0.989	1.041
first_careunitCoronary Care Unit (CCU)	0.932	0.025	-2.819	0.0048	0.888	0.979
first_careunitIntensive Care Unit (ICU)	0.000	5497.742	-0.005	0.9957	0.000	Inf
first_careunitMed/Surg	0.582	1.961	-0.276	0.7826	0.012	27.188
first_careunitMedical Intensive Care Unit (MICU)	0.956	0.021	-2.122	0.0338	0.917	0.997
first_careunitMedical/Surgical Intensive Care Unit (MICU/SICU)	1.423	0.023	15.502	0.0000	1.361	1.487
first_careunitMedicine	0.000	9241.318	-0.003	0.9974	0.000	Inf

Predictor	IRR	Std. Error	Statistic	p-value	95% CI Low	95% CI High
first_careunitMedicine/Cardiology Intermediate	0.000	42247.166	-0.001	0.9995	0.000	Inf
first_careunitNeuro Intermediate	0.164	0.031	-59.097	0.0000	0.154	0.174
first_careunitNeuro Stepdown	0.486	0.055	-13.226	0.0000	0.437	0.541
first_careunitNeuro Surgical Intensive Care Unit (Neuro SICU)	0.546	0.050	-12.182	0.0000	0.496	0.602
first_careunitNeurology	0.192	1.960	-0.842	0.3997	0.004	8.942
first_careunitPACU	0.004	0.196	-28.138	0.0000	0.003	0.006
first_careunitSurgery/Trauma	0.000	11666.393	-0.003	0.9980	0.000	Inf
first_careunitSurgery/Vascular/Intermediate	0.000	2927.740	-0.010	0.9918	0.000	Inf
first_careunitSurgical Intensive Care Unit (SICU)	1.038	0.024	1.592	0.1115	0.991	1.088
first_careunitTrauma SICU (TSICU)	0.878	0.025	-5.144	0.0000	0.836	0.923

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Table 9. Model 3 — Total CDS Events per Day

Predictor	IRR	Std. Error	Statistic	p-value	95% CI Low	95% CI High
(Intercept)	394.235	0.019	309.792	0.0000	379.606	409.429
anchor_age	0.998	0.000	-10.534	0.0000	0.997	0.998
genderM	1.035	0.008	4.416	0.0000	1.019	1.051
first_careunitCoronary Care Unit (CCU)	0.859	0.015	-10.120	0.0000	0.834	0.885
first_careunitIntensive Care Unit (ICU)	0.144	0.207	-9.361	0.0000	0.096	0.216
first_careunitMed/Surg	0.550	1.186	-0.505	0.6137	0.054	5.614
first_careunitMedical Intensive Care Unit (MICU)	0.855	0.013	-12.192	0.0000	0.834	0.877
first_careunitMedical/Surgical Intensive Care Unit (MICU/SICU)	1.202	0.014	13.374	0.0000	1.170	1.235
first_careunitMedicine	0.067	0.297	-9.116	0.0000	0.037	0.120
first_careunitMedicine/Cardiology Intermediate	0.820	1.185	-0.167	0.8673	0.080	8.362
first_careunitNeuro Intermediate	0.189	0.019	-89.981	0.0000	0.182	0.196
first_careunitNeuro Stepdown	0.418	0.033	-26.397	0.0000	0.392	0.446
first_careunitNeuro Surgical Intensive Care Unit (Neuro SICU)	0.522	0.030	-21.654	0.0000	0.492	0.554
first_careunitNeurology	0.160	1.184	-1.547	0.1218	0.016	1.630
first_careunitPACU	1.659	0.108	4.699	0.0000	1.343	2.050
first_careunitSurgery/Trauma	0.096	0.375	-6.250	0.0000	0.046	0.200
first_careunitSurgery/Vascular/Intermediate	0.096	0.099	-23.715	0.0000	0.079	0.116
first_careunitSurgical Intensive Care Unit (SICU)	0.933	0.014	-4.861	0.0000	0.907	0.959

Predictor	IRR	Std. Error	Statistic	p-value	95% CI Low	95% CI High
first_careunit Trauma SICU (TSICU)	0.852	0.015	-10.540	0.0000	0.827	0.878

Visualizations

Distribution of Total CDS-Triggering Events

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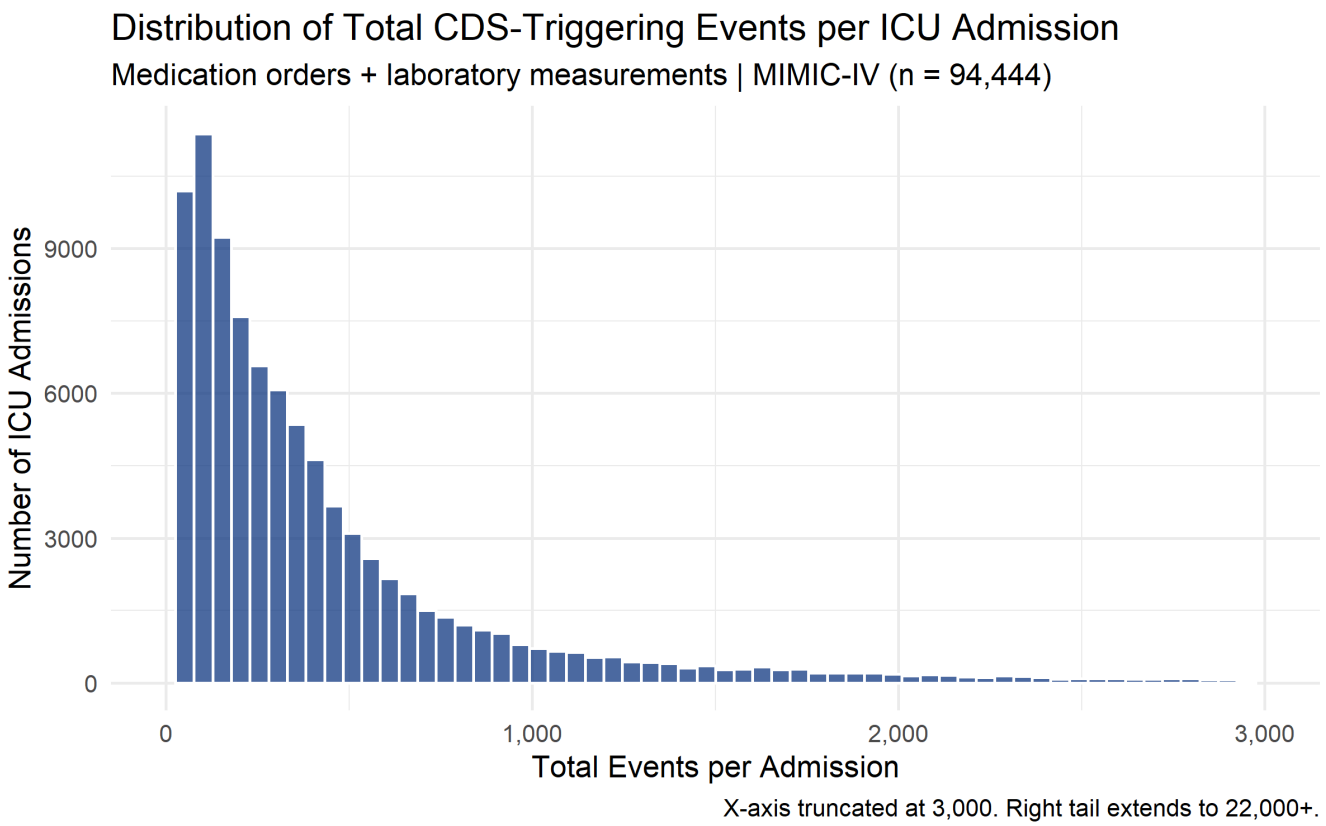

Figure 2. Distribution of total CDS-triggering events per ICU admission. Medication orders and laboratory measurements combined. X-axis truncated at 3,000; right tail extends to 22,000+. MIMIC-IV, n = 94,444 ICU stays.

Event Rate by ICU Unit Type

Show

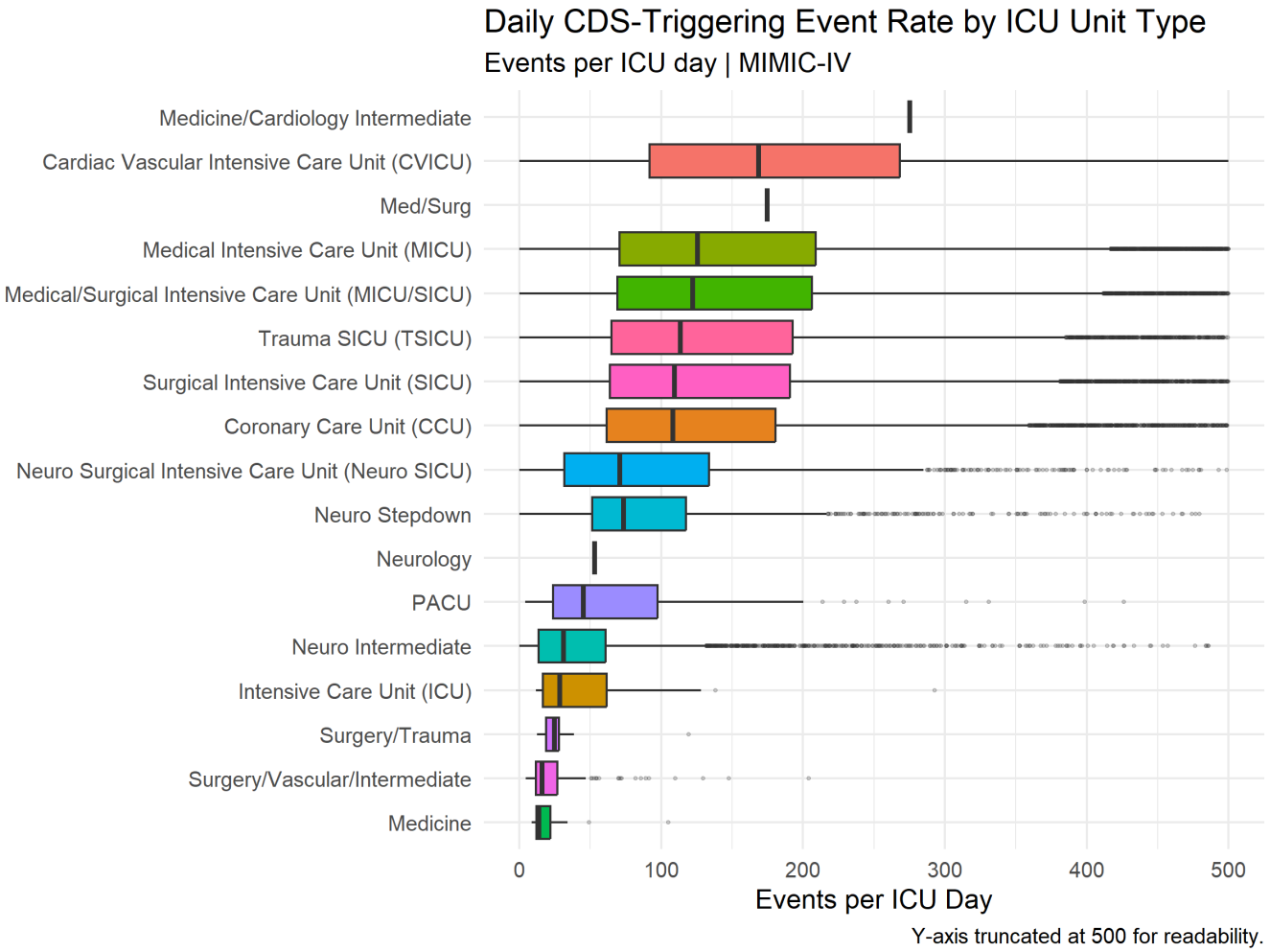

Figure 3. Daily CDS-triggering event rate by ICU unit type. Events per ICU day (medication orders + lab measurements). Y-axis truncated at 500 for readability. MIMIC-IV.

Medication Orders vs. Lab Events

Show

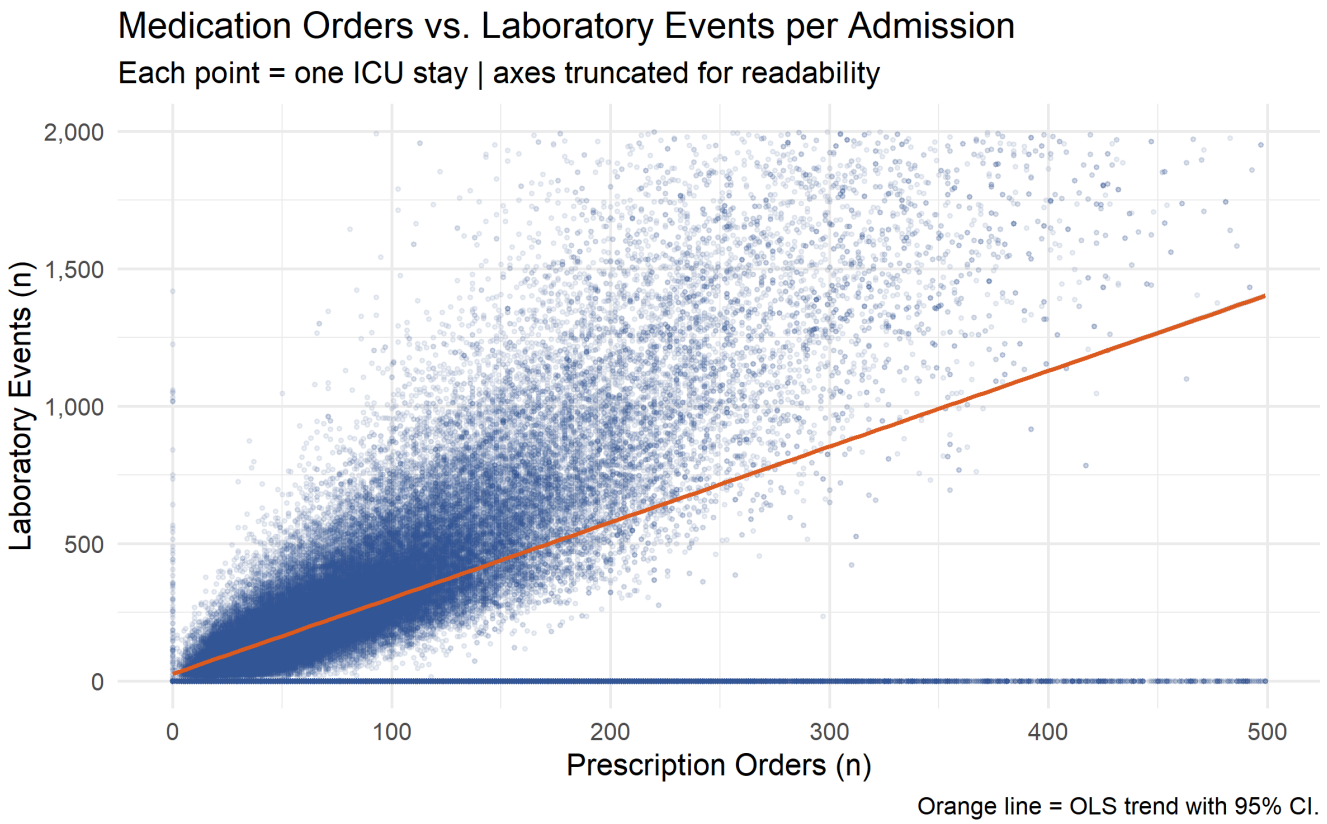

Figure 4. Relationship between medication orders and laboratory events per ICU admission. Each point represents one ICU stay. Axes truncated for readability. Orange line shows OLS trend with 95% confidence interval. MIMIC-IV.

Suppression Scenario Comparison

Show

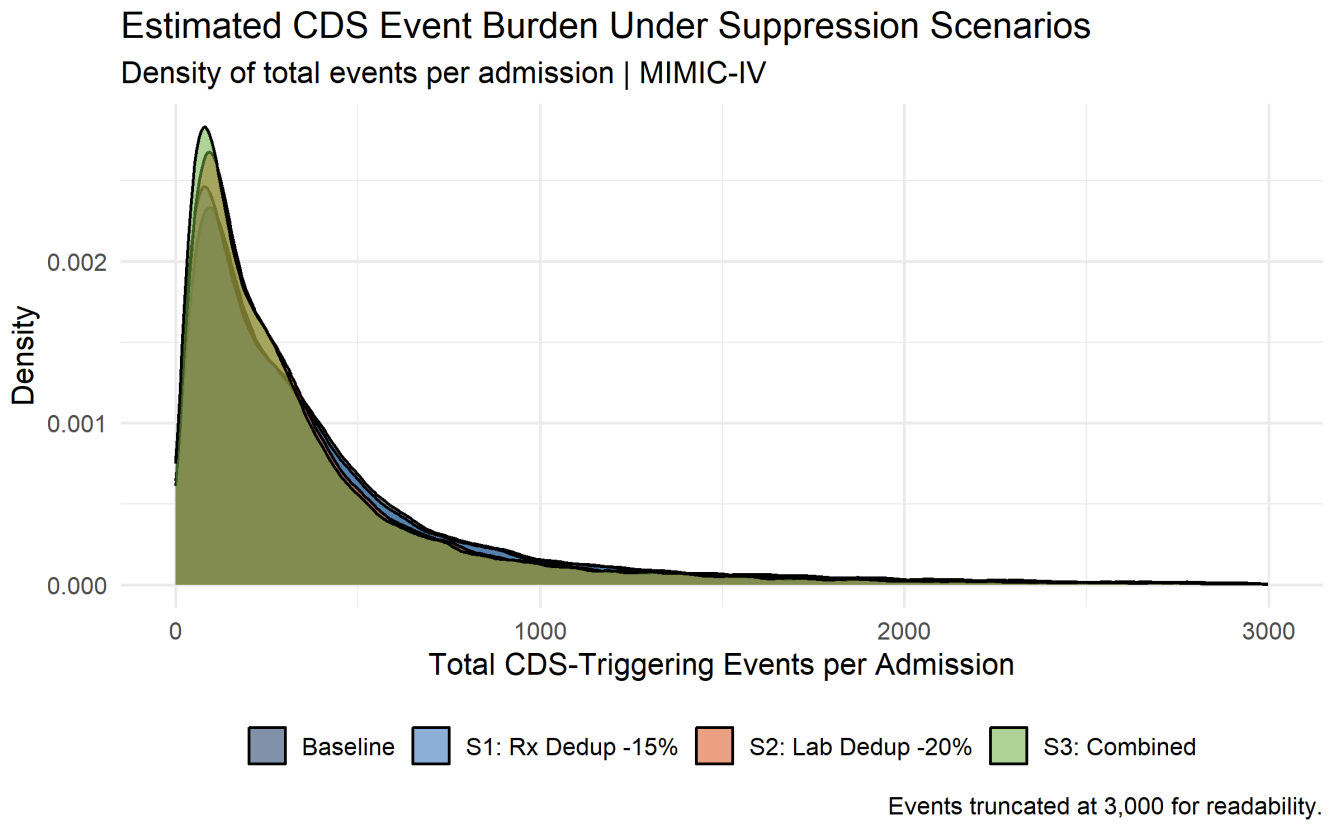

Figure 5. Estimated CDS event burden under three suppression scenarios compared to baseline. S1 = medication order deduplication (-15%); S2 = laboratory event suppression (-20%); S3 = combined strategy. Events truncated at 3,000 for readability. MIMIC-IV, n = 94,444.

ICU Length of Stay Distribution

Show

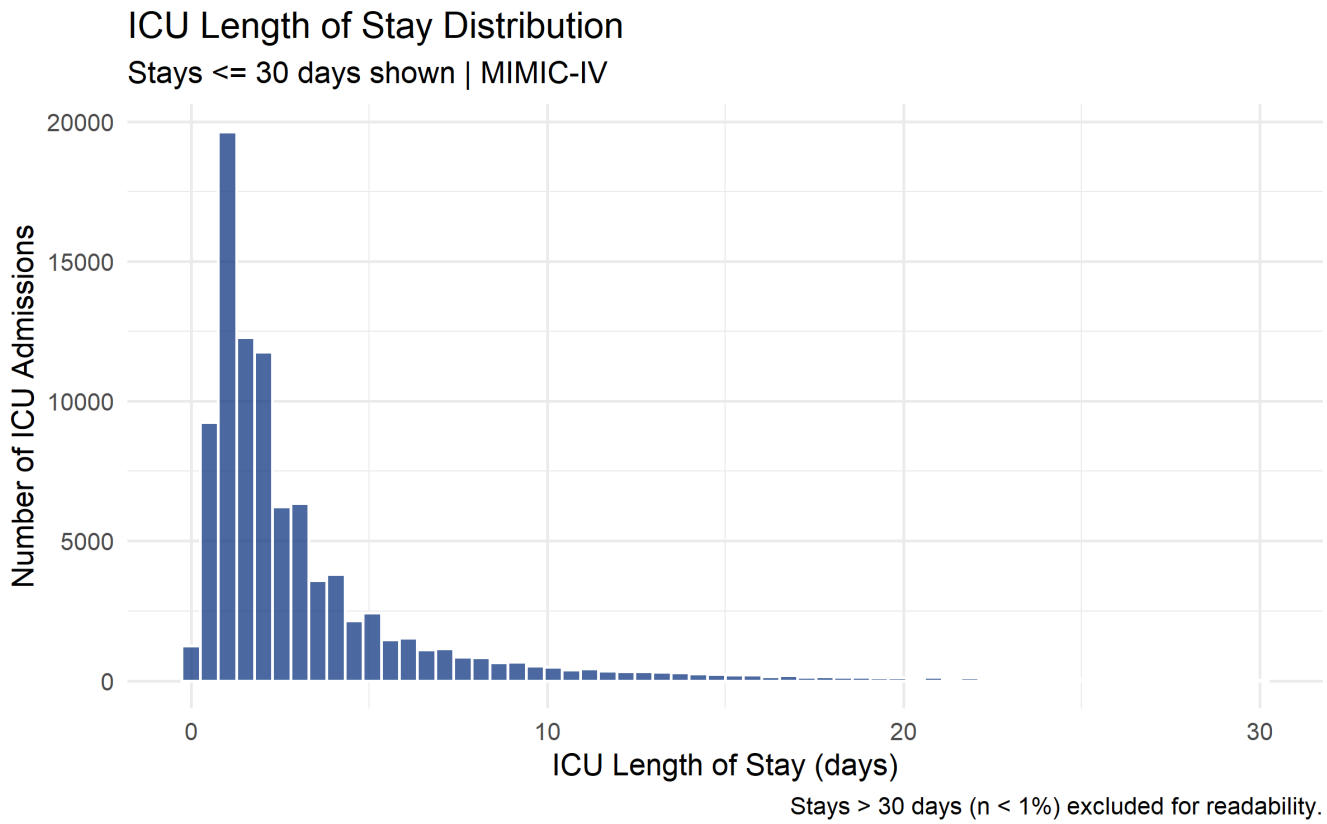

Figure 6. ICU length of stay distribution. Stays exceeding 30 days (less than 1% of cohort) excluded for readability. MIMIC-IV, n = 94,444.

Temporal Clustering Visualizations

Medication Orders by Hour of Day

Show

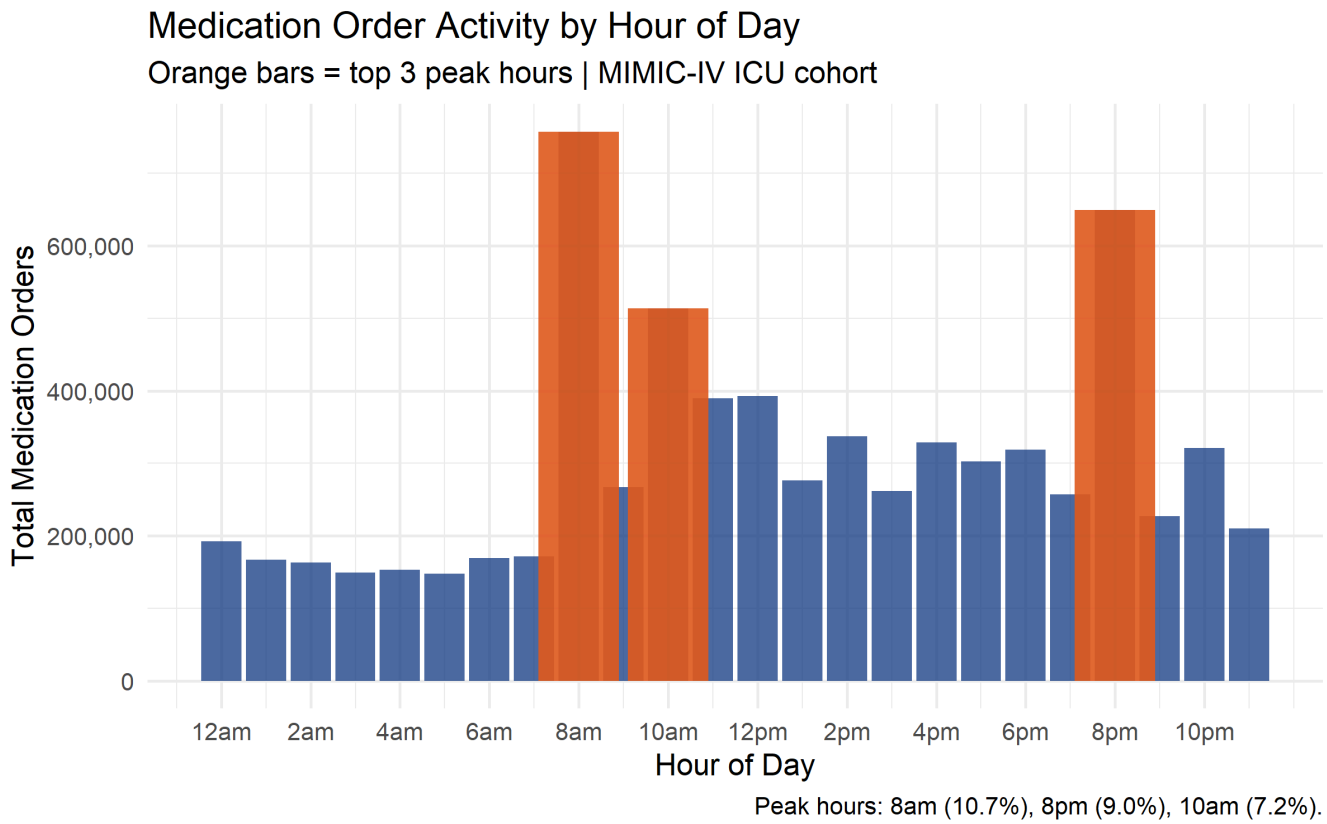

Figure 7. Medication order activity by hour of day across all ICU cohort admissions. Orange bars indicate the three peak hours: 8am (10.7%), 8pm (9.0%), and 10am (7.2%). MIMIC-IV, 8,578,568 total orders.

Orders by Clinical Shift

Show

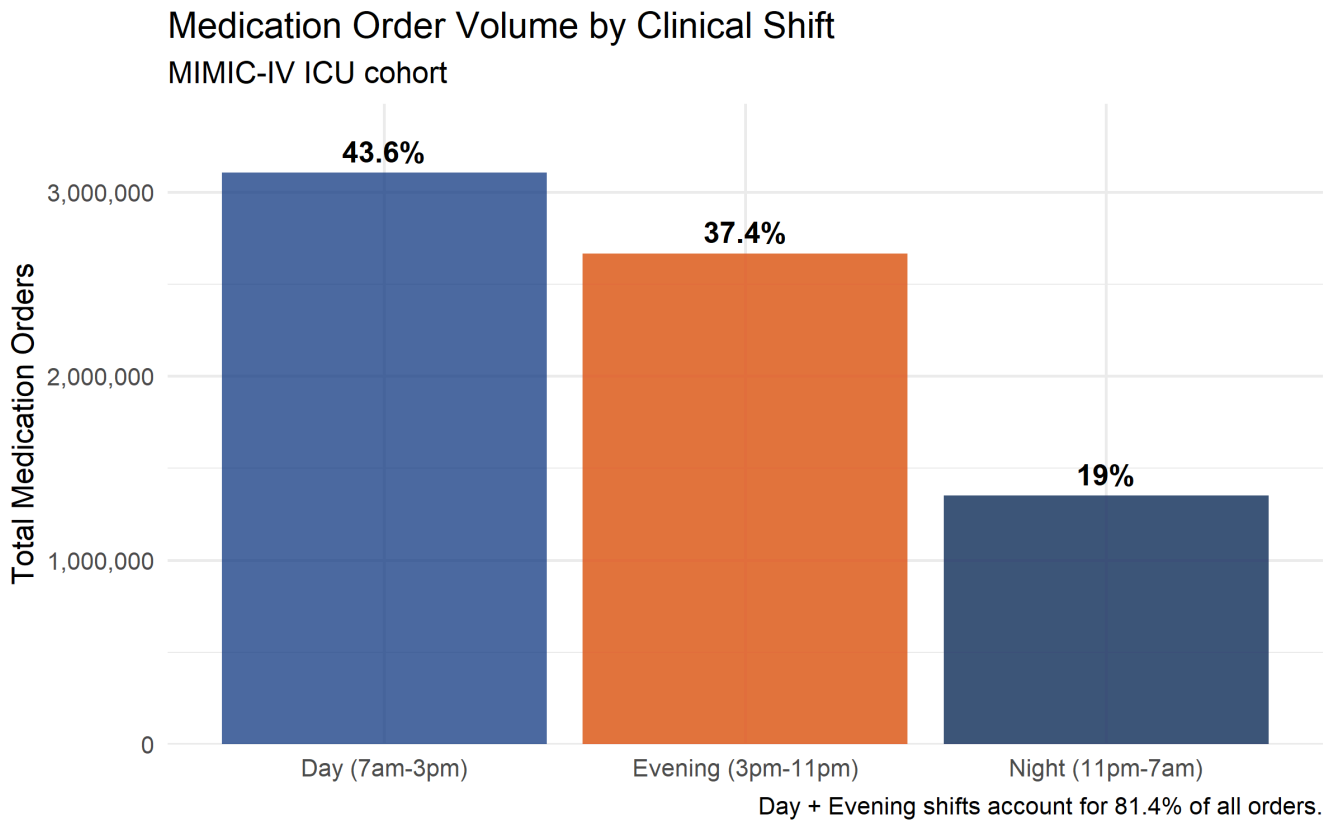

Figure 8. Medication order volume by clinical shift. Day and evening shifts together account for 81.4% of all orders. MIMIC-IV ICU cohort.

Orders by Day of ICU Stay

Show

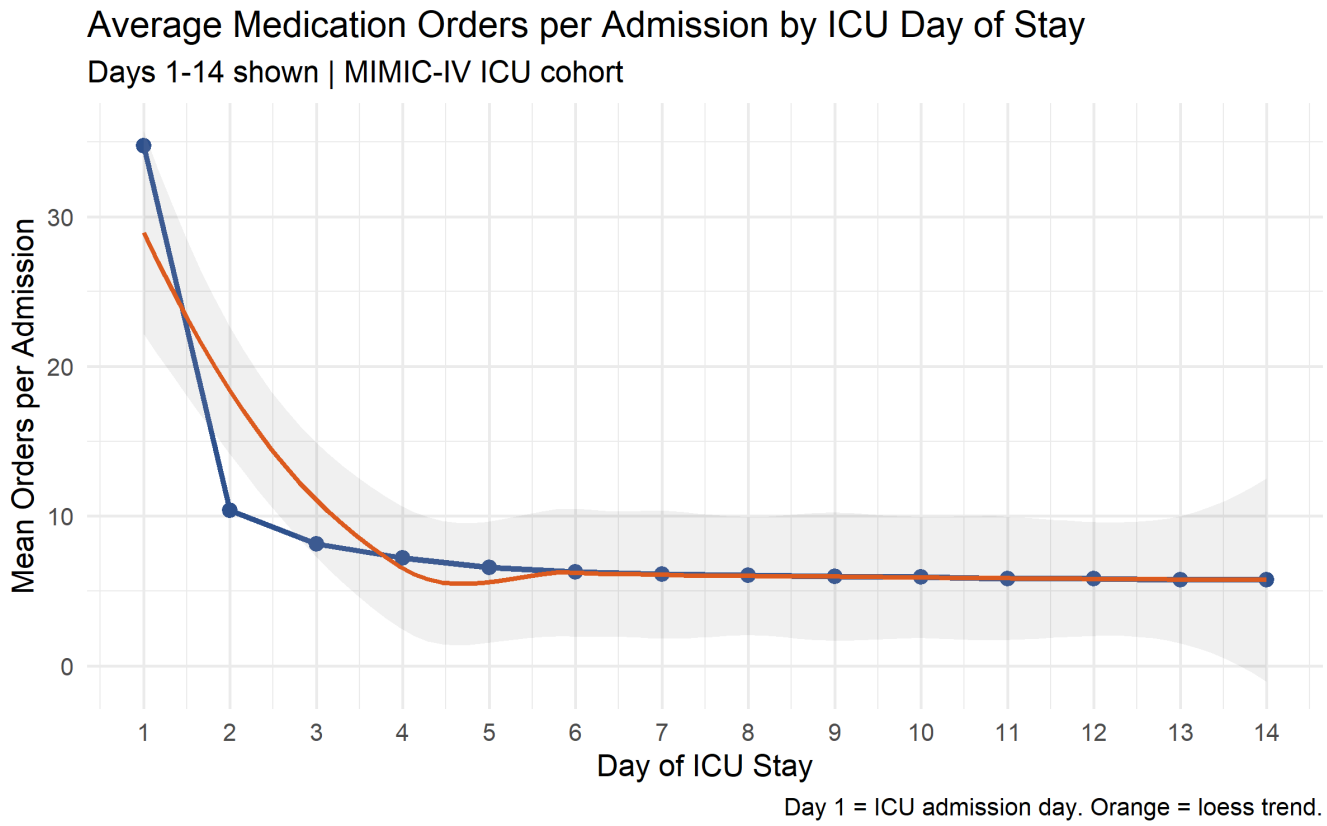

Figure 9. Average medication orders per admission by ICU day of stay. Days 1 through 14 shown. Day 1 corresponds to ICU admission day. Orange line shows loess smoothing trend. MIMIC-IV ICU cohort.

60-Minute Burst Distribution

Show

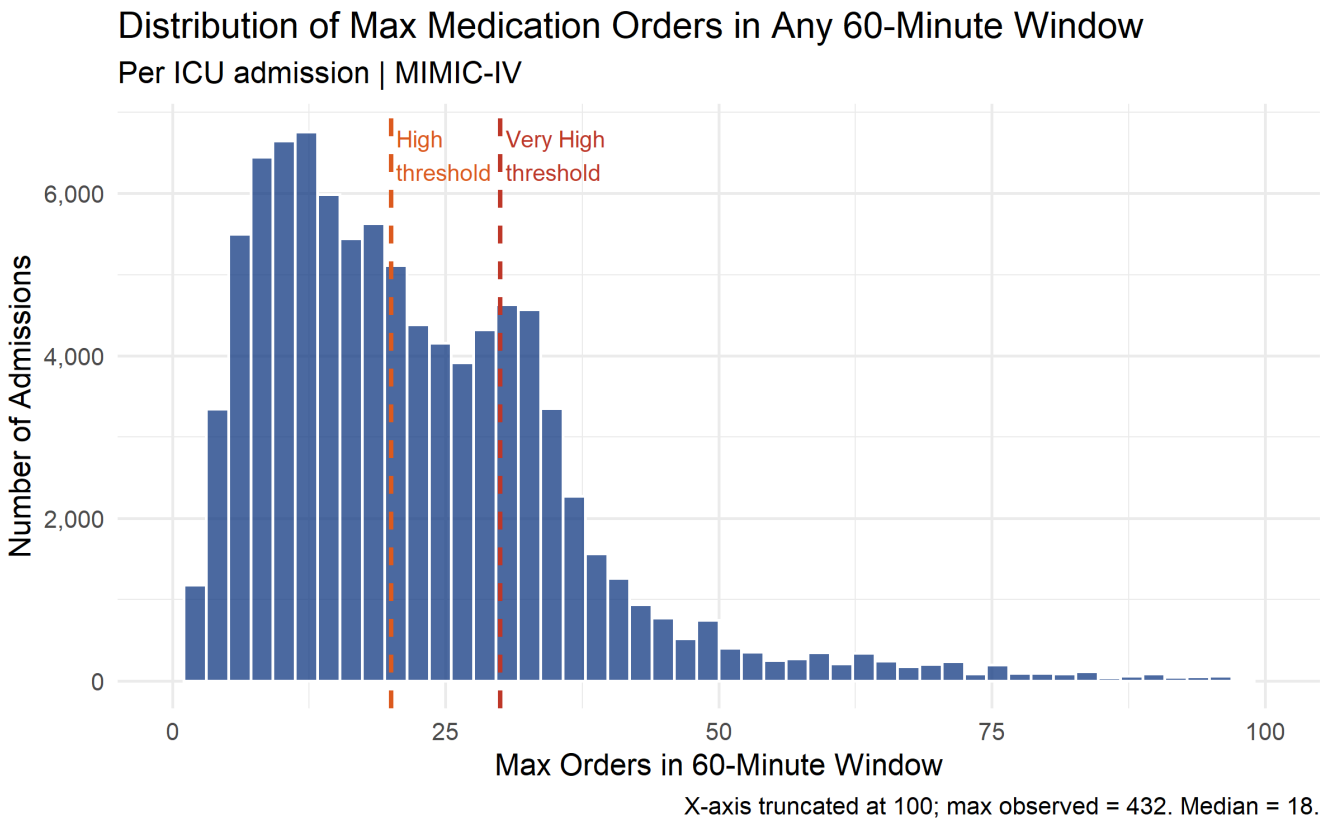

Figure 10. Distribution of maximum medication orders placed within any 60-minute window per ICU admission. Dashed lines mark High (20+) and Very High (30+) thresholds. X-axis truncated at 100; maximum observed value was 432. Median = 18. MIMIC-IV.

Burst Intensity by Category

Show

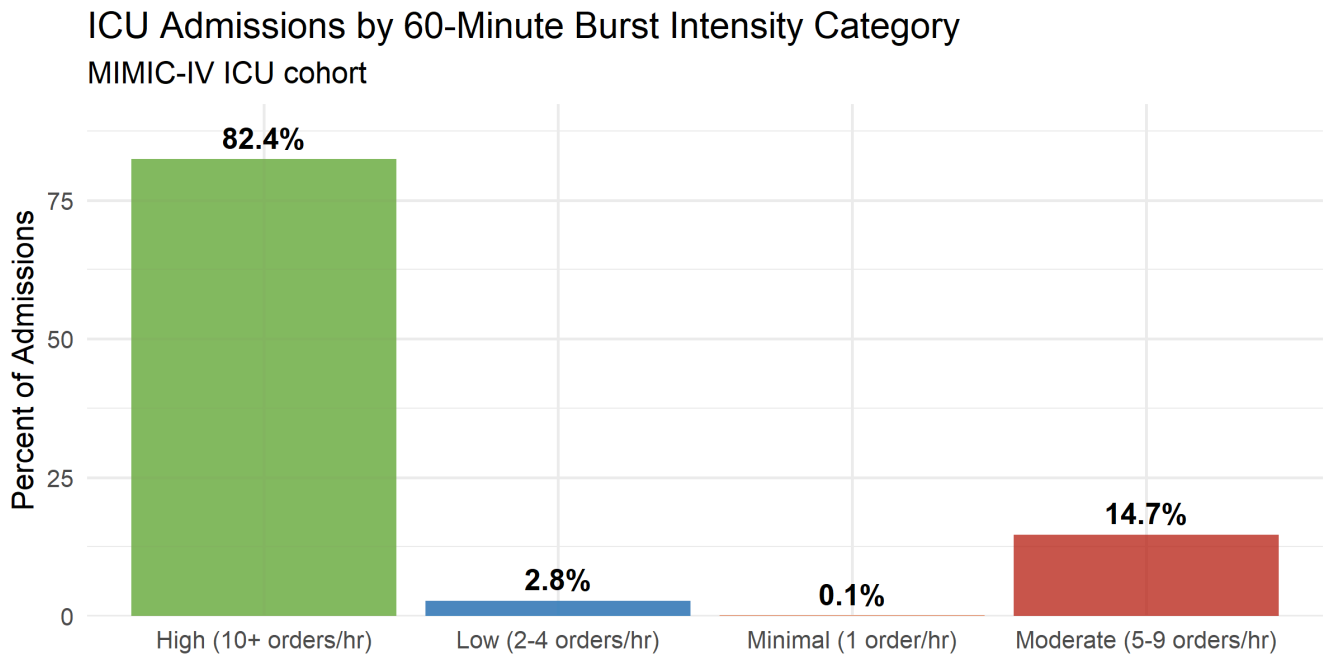

Figure 11. Percentage of ICU admissions by 60-minute burst intensity category. Thresholds: Low <10, Moderate 10-19, High 20-29, Very High 30+ orders per hour. MIMIC-IV.

Preprocessing Notes

The following decisions were made during data preparation. Each is documented with the reason it was made so the analysis can be understood and reproduced.

Lab events queried via DuckDB — The labevents file in MIMIC-IV contains tens of millions of rows, exceeding available RAM. DuckDB was used to run SQL queries directly on disk, pulling only rows belonging to patients in the ICU cohort without loading the full file into memory.

Prescriptions loaded with `select = c(...)` — Only the five columns needed for analysis were loaded from the prescriptions file, reducing RAM usage by approximately 80% compared to loading all 26 columns.

Prescription counts aggregated before joining — Medication orders were summarized at the admission level before merging with the ICU cohort. Joining raw prescription records directly would have multiplied the number of rows in the dataset by hundreds, making it unmanageable.

ICU units with n < 200 collapsed to "Other" — Eight ICU unit types had fewer than 200 stays. Units this small cause convergence failure in negative binomial estimation because the model cannot reliably estimate coefficients from so few observations.

Lab counts winsorized at the 99th percentile (3,086) — The maximum lab event count was 19,824. Values this extreme caused the negative binomial model to fail. Winsorizing replaced the top 1% of values with the 99th percentile cap, leaving 99% of the data unchanged.

Rate variables derived (events per ICU day) — Raw counts are confounded by length of stay. A patient staying 30 days accumulates more events than a 2-day patient regardless of care intensity. Dividing by ICU days produces a fair per-day rate for comparison.

Negative binomial over Poisson — Confirmed by likelihood ratio test: Chi-sq = 8,224,010, p ≈ 0. Variance-to-mean ratios of 119 to 1,178 across outcomes confirm severe overdispersion (see Table 4 and Table 5).

Cohort lookup deduplicated before temporal join — Some hospital admissions contain multiple ICU stays, giving multiple intime values per hadm_id. A deduplicated lookup table was created before joining prescription timestamps to prevent row multiplication in the temporal analysis.

Burst thresholds recalibrated to 10/20/30 — The original thresholds (2/5/10) classified 80.7% of admissions as High because the median burst in this dataset is 18 orders per hour. Recalibrated thresholds produce a meaningful spread across all four categories (see Figure 11).

Show

```
##   mean_rx median_rx sd_rx mean_rxpd median_rxpd
## 1   113.8        83 116.2      77.6        35.8
```

Show

```
##   mean_lab median_lab sd_lab mean_labpd median_labpd
## 1    380.6        198  669.4      253.2           91
```

Show

```
##   mean_total median_total mean_totalpd median_totalpd
## 1      494.4          280        330.8          126.9
```